

Feature Importance using random forest and Decision Tree

- all the ^(columns) features aren't imp. [colⁿ]
- selecting and removing unnecessary features is imp.

[rf.feature_importances_]

sns.heatmap(rf.feature_importances_.reshape(28, 28))

- Feature Importance in Decision Tree.

$$f_{ik} = \frac{\sum_{j \in \text{node split on feature } k} \text{node importance}}{\sum_{j \in \text{all node}} \text{node importance}}$$

find feature imp of decision tree and do average, then we get the feature importance of random forest.